

Data Management

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```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
v dplyr      1.1.4      v readr      2.1.5
```

```
v forcats    1.0.0      v stringr    1.5.1
```

```
v ggplot2     3.5.1      v tibble     3.2.1
```

```
v lubridate  1.9.4      v tidyr      1.3.1
```

```
v purrr       1.0.4
```

```
-- Conflicts ----- tidyverse_conflicts() --
```

```
x dplyr::filter() masks stats::filter()
```

```
x dplyr::lag()     masks stats::lag()
```

```
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(descr)
```

```
library(knitr)
```

```
library(dplyr)
```

```
library(readr)
```

```
library(readxl)
```

```
library(ggplot2)
```

```
===My Data===
```

```
file_path <- "/cloud/project/data/data.xls"
```

```
data.xls <- read_excel(file_path)
```

```
options(width=80)
```

```
head(data.xls)
```

```
# A tibble: 6 x 138
```

	RespondentID	Country	A3.1	A3.2	A3.3	A3.4	A3.5	A3.6	A3.7	A3.8	A3.9
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	1	1	2	3	52	52	4	1	5	12	1

```

2          2          1          2          5          52          52          3          1          8          2          5
3          3          1          1          4          94          94          4          1          5          2          1
4          4          1          1          2          221          221          1          1          10          3          4
5          5          1          1          3          52          52          3          1          8          2          0
6          6          1          1          1          52          52          1          4          9          3          4
# i 127 more variables: A3.10 <dbl>, A5.1 <dbl>, A5.2 <dbl>, `A5.3*` <dbl>,
#   A5.4 <dbl>, A5.5 <dbl>, A6.1 <dbl>, A7.1 <dbl>, A7.2 <dbl>, A9.1 <dbl>,
#   A9.2 <dbl>, `A9.3*` <dbl>, A9.4 <dbl>, A9.5 <dbl>, A9.6 <dbl>, A9.7 <dbl>,
#   `A9.8*` <dbl>, A11.1 <dbl>, A11.2 <dbl>, A11.3 <dbl>, A11.4 <dbl>,
#   A12.1 <dbl>, A12.2 <dbl>, A12.3 <dbl>, A12.4 <dbl>, A13.1 <dbl>,
#   A13.2 <dbl>, A13.3 <dbl>, A13.4 <dbl>, A14.1 <dbl>, A14.2 <dbl>,
#   A14.3 <dbl>, A14.4 <dbl>, A16.1 <dbl>, A16.2 <dbl>, `A16.3*` <dbl>, ...

```

===Frequency===

```
freq(as.ordered(data.xls$A3.8), plot = FALSE)
```

```

as.ordered(data.xls$A3.8)
  Frequency  Percent Cum Percent
1         293  14.4264    14.43
2         566  27.8680    42.29
3          87   4.2836    46.58
4          22   1.0832    47.66
5          11   0.5416    48.20
6          10   0.4924    48.70
7          11   0.5416    49.24
8          15   0.7386    49.98
9          55   2.7080    52.68
10         155   7.6317    60.32
11          75   3.6928    64.01
12         731  35.9921   100.00
Total      2031 100.0000

```

===Frequency of Religious Orientations===

```
freq(as.ordered(data.xls$A3.8), plot = FALSE)
```

```

as.ordered(data.xls$A3.8)
  Frequency  Percent Cum Percent
1         293  14.4264    14.43
2         566  27.8680    42.29
3          87   4.2836    46.58
4          22   1.0832    47.66
5          11   0.5416    48.20
6          10   0.4924    48.70
7          11   0.5416    49.24

```

8	15	0.7386	49.98
9	55	2.7080	52.68
10	155	7.6317	60.32
11	75	3.6928	64.01
12	731	35.9921	100.00
Total	2031	100.0000	

===Frequency of Levels of Educations===

```
freq(as.ordered(data.xls$A3.10), plot = FALSE)
```

```
as.ordered(data.xls$A3.10)
```

	Frequency	Percent	Cum Percent
1	49	2.413	2.413
2	323	15.903	18.316
3	330	16.248	34.564
4	565	27.819	62.383
5	441	21.713	84.097
6	252	12.408	96.504
7	37	1.822	98.326
8	34	1.674	100.000
Total	2031	100.000	

This frequency table shows the levels of education within a sample. It shows me how many people don't have a high school degree to how many people have a doctorates. This will help me relate the info to people who are religious, to their levels of education. I need this info because I will eventually compare the information I gather to how many took COVID-19 as a serious threat.

===Frequency of who believed COVID-19 was exaggerated===

```
freq(as.ordered(data.xls$`A5.3*`), plot = FALSE)
```

```
as.ordered(data.xls$`A5.3*`)
```

	Frequency	Percent	Cum Percent
1	592	29.148	29.15
2	401	19.744	48.89
3	259	12.752	61.64
4	168	8.272	69.92
5	148	7.287	77.20
6	139	6.844	84.05
7	122	6.007	90.05
8	94	4.628	94.68
9	60	2.954	97.64
10	48	2.363	100.00
Total	2031	100.000	

This frequency table shows the levels in which how many people thought COVID-19 was being exaggerated. It goes from level 1 = Not at all to level 10= Extremely. This will help me see how seriously people took COVID-19 and who underestimated it.

===Frequency in which mask was worn===

```
freq(as.ordered(data.xls$A31.4), plot = FALSE)
```

```
as.ordered(data.xls$A31.4)
      Frequency Percent Cum Percent
1           874  43.033      43.03
2           121   5.958      48.99
3           120   5.908      54.90
4           185   9.109      64.01
5           184   9.060      73.07
6           195   9.601      82.67
7           352  17.331     100.00
Total        2031 100.000
```

This frequency table shows how many people had frequently put on a mask during COVID-19. It goes from level 1, Never, to level 7, Always. This will help aid my question on who was taking COVID-19 seriously.

===Distance Away Frequency===

```
freq(as.ordered(data.xls$A31.6), plot = FALSE)
```

```
as.ordered(data.xls$A31.6)
      Frequency Percent Cum Percent
1             24   1.1817      1.182
2             20   0.9847      2.166
3             36   1.7725      3.939
4            106   5.2191      9.158
5            138   6.7947     15.953
6            496  24.4215     40.374
7           1211  59.6258     100.000
Total        2031 100.0000
```

This table shows the frequency in which a person kept a distance of 1.5 meters away from others. The chart goes from level 1, Never, to level 7, Always. This will help me see who didn't wear their masks during COVID-19, which relates to my question of who didn't follow the guidelines provided to everyone across the globe.

=== Variable for Levels of Education===

```
data.xls$LVL_EDU <- factor(ifelse(data.xls$A3.10 %in% c("1","2"), "High School and Below", "College and Above"))
summary(data.xls$LVL_EDU)
```

High School and Below	College and Above
1659	372

I am creating a variable named LVL_EDU to help me see who have educations from up to High School then College and higher.

===Variable for People of Only Religious Orientations===

```
data.xls$ONLY_R <- factor(ifelse(data.xls$A3.8 %in% c("1", "2","3","4","5","6","7","8","9","10"), "Has a Religious Belief", "No Beliefs"))
summary(data.xls$ONLY_R)
```

Has a Religious Belief	No Beliefs
1145	886

I am creating a variable named ONLY_R to group up who and who don't have a religious orientation.

===Variable for People Who Think COVID-19 was over exaggerated===

```
data.xls$CVD_EXAG <- factor(ifelse(data.xls$`A5.3*` %in% c("1", "2","3","4","5"), "Reasonable Concern", "Out-of-Proportion Concern"))
summary(data.xls$CVD_EXAG)
```

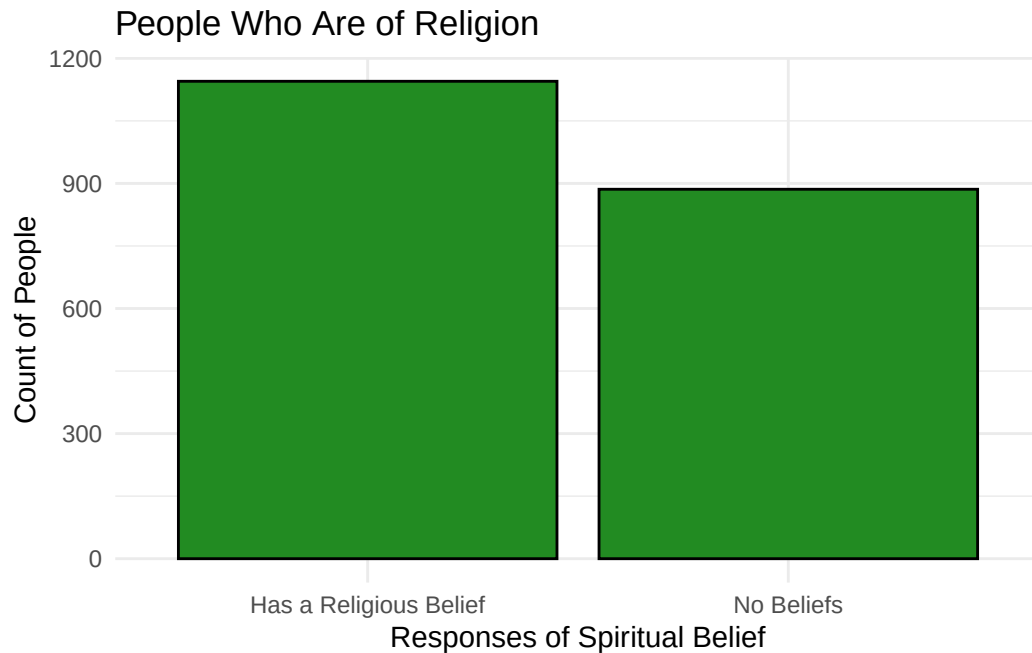
Reasonable Concern	Out-of-Proportion Concern
1568	463

I am creating the variable CVD_EXAG to summarize the responses 1 = Not at all, 10= Extremely into two managed variables, options 1-5 being considered the people who believed that COVID-19 was not being over-exaggerated and options 6-10 being considered the people who thought the virus was being told to be a larger issue than it really was.

===Univariate Graphs===

```
library(ggplot2)

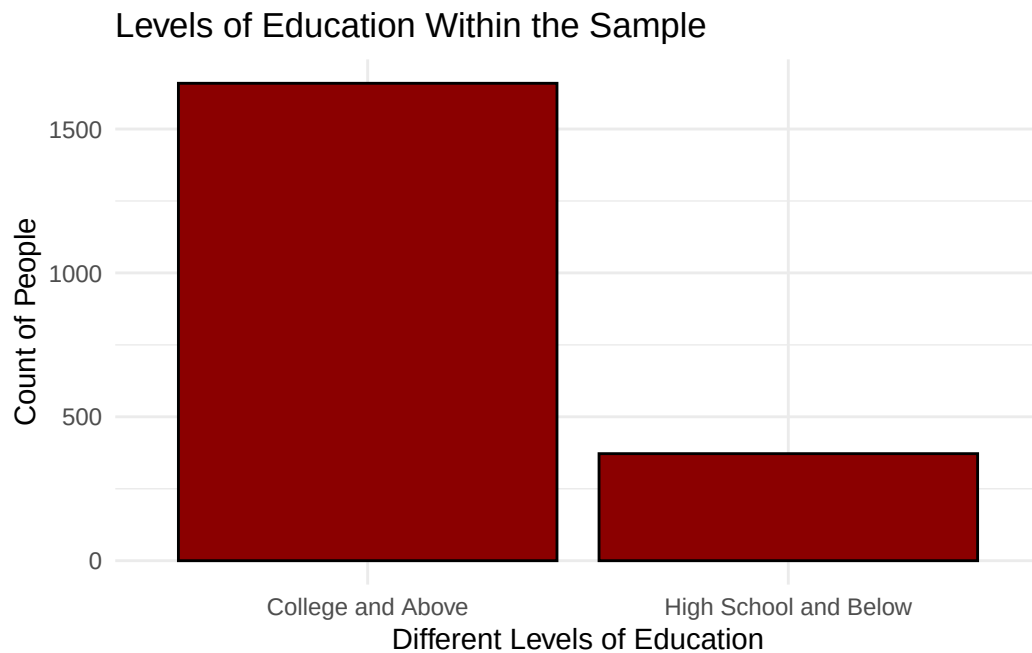
data.xls %>%
  ggplot(aes(x = ONLY_R)) +
  geom_bar(fill = "forestgreen", color = "black") +
  ggtitle("People Who Are of Religion") +
  xlab("Responses of Spiritual Belief") +
  ylab("Count of People") +
  theme_minimal()
```



I am developing this graph to show how many people in the sample do and do not have any religious perspectives. I would like to correlate this graph later on with my other variables to see if there are any patterns between each other. One of my main concerns is that some people of religious backgrounds didn't think COVID-19 was a problem and that nothing would happen to each other. With this data gathered and graphed, I think it'll become easier to start comparing.

```
library(ggplot2)

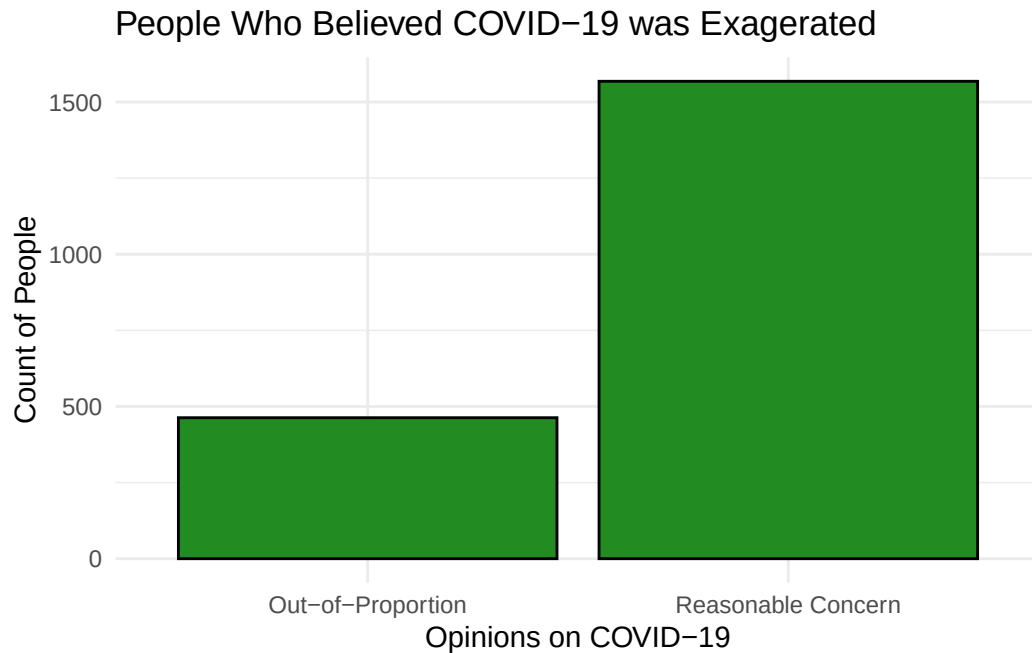
data.xls %>%
  ggplot(aes(x = LVL_EDU)) +
  geom_bar(fill = "darkred", color = "black") +
  ggtitle("Levels of Education Within the Sample") +
  xlab("Different Levels of Education") +
  ylab("Count of People") +
  theme_minimal()
```



This graph will help to visualize the differences between two major levels of education. The goal of this is to understand whether or not the levels education dictate what people believed during the COVID-19 pandemic and how serious the virus was. My graph here is just to show how many in the sample have received higher education or not.

```
library(ggplot2)

data.xls %>%
  ggplot(aes(x = CVD_EXAG)) +
  geom_bar(fill = "forestgreen", color = "black") +
  ggtitle("People Who Believed COVID-19 was Exaggerated") +
  xlab("Opinions on COVID-19") +
  ylab("Count of People") +
  theme_minimal()
```



This graph shows the many people who believed whether COVID-19 was overtly told as a massive issue and the many others who believed the concern of the virus was authentic. The main goal for this chart is to get an idea of the opinions many had about the virus when the issue was most prevalent.

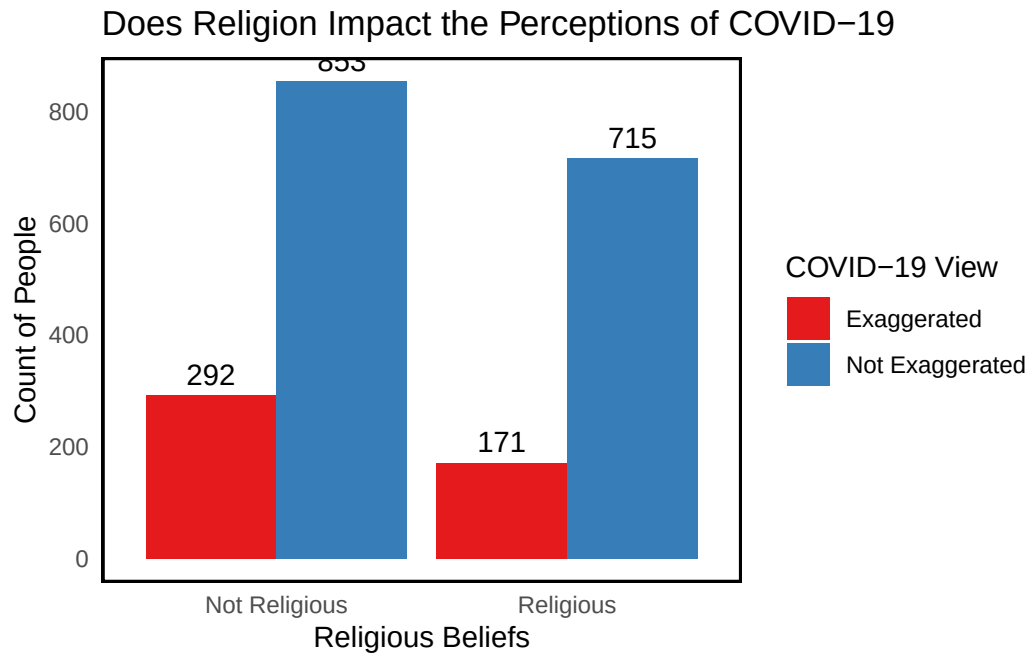
====Bivariate Graph=====

====Is Religion a Cause of Disregarding the COVID-19 Virus?=====

```
ggplot(data=data.xls, aes(x = factor(ONLY_R, labels = c("Not Religious", "Religious")), fill =
  geom_bar(position = "dodge") +
  geom_text(stat='count', aes(label=..count..), position=position_dodge(0.9), vjust=-0.5) +
  labs(title = "Does Religion Impact the Perceptions of COVID-19",
    x = "Religious Beliefs",
    y = "Count of People",
    fill = "COVID-19 View") +
  theme_minimal() +
  scale_fill_brewer(palette = "Set1") +
  theme(panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.border = element_rect(color = "black", fill = NA, size = 1))
```

Warning: The `size` argument of `element\_rect()` is deprecated as of ggplot2 3.4.0.
i Please use the `linewidth` argument instead.

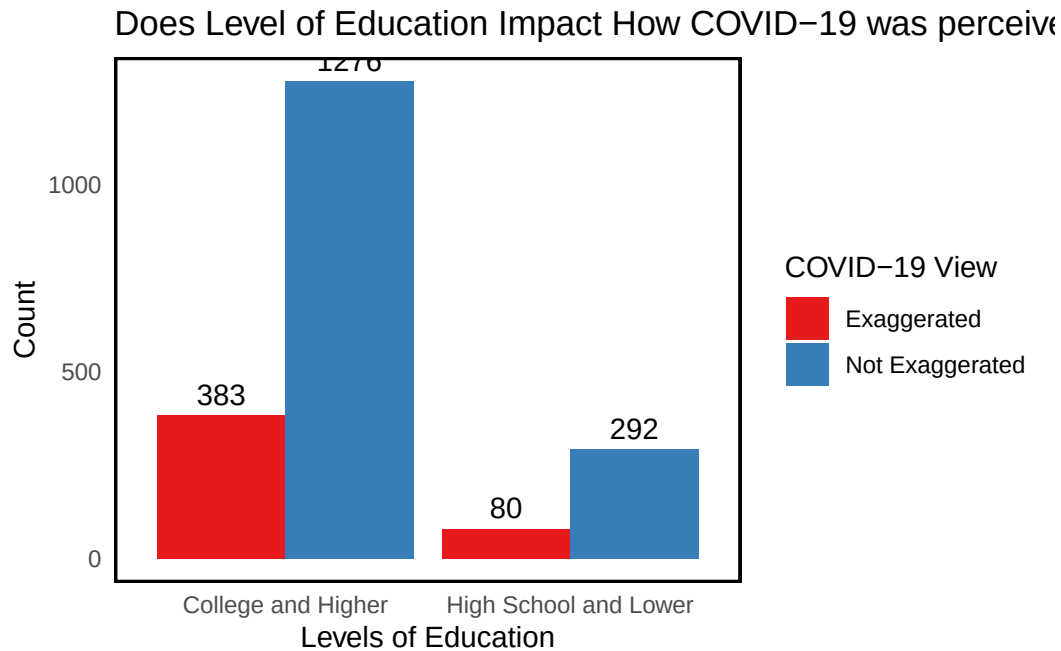
Warning: The dot-dot notation (`..count..`) was deprecated in ggplot2 3.4.0.
i Please use `after\_stat(count)` instead.



Through my findings I have found out that regardless of religion, a majority within a sample have agreed that COVID-19 was as serious as it was. I wanted to make this graph because through my own experiences during the pandemic, I witnessed many people disregarding the virus and explaining that their place of worship denied the virus being real or even an issue. This graph helps me see that it isn't really like this everywhere else.

===Did the Level of Education Affect the Perception of COVID-19 as a serious issue?===

```
ggplot(data=data.xls, aes(x = factor(LVL_EDU, labels = c("College and Higher", "High School and below"),
  geom_bar(position = "dodge") +
  geom_text(stat='count', aes(label=..count..), position=position_dodge(0.9), vjust=-0.5) +
  labs(title = "Does Level of Education Impact How COVID-19 was perceived?",
    x = "Levels of Education",
    y = "Count",
    fill = "COVID-19 View") +
  theme_minimal() +
  scale_fill_brewer(palette = "Set1") +
  theme(panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.border = element_rect(color = "black", fill = NA, size = 1))
```



A common issue that's been witnessed among many is how a lack of education can lead people to be ignorant about certain theories, ideas or health issues. One of the most leading issues would be people who practice anti-vaxxing, the act of refusing to use a vaccine for medical problems. The cause of this is usually the lack of education to help a person understand why they should be taking these medical solutions. This is one of the biggest barriers in vaccine uptake. I wanted to see if the differences in education within the sample saw COVID-19 as an issue or not. Through my research and the data provided we can see that many, highly educated or not, did see the virus as an issue to take seriously.

===Chi-Square Test===

```
table(data.xls$ONLY_R)
```

Has a Religious Belief	No Beliefs
1145	886

```
chisq.test(table(data.xls$ONLY_R))$expected
```

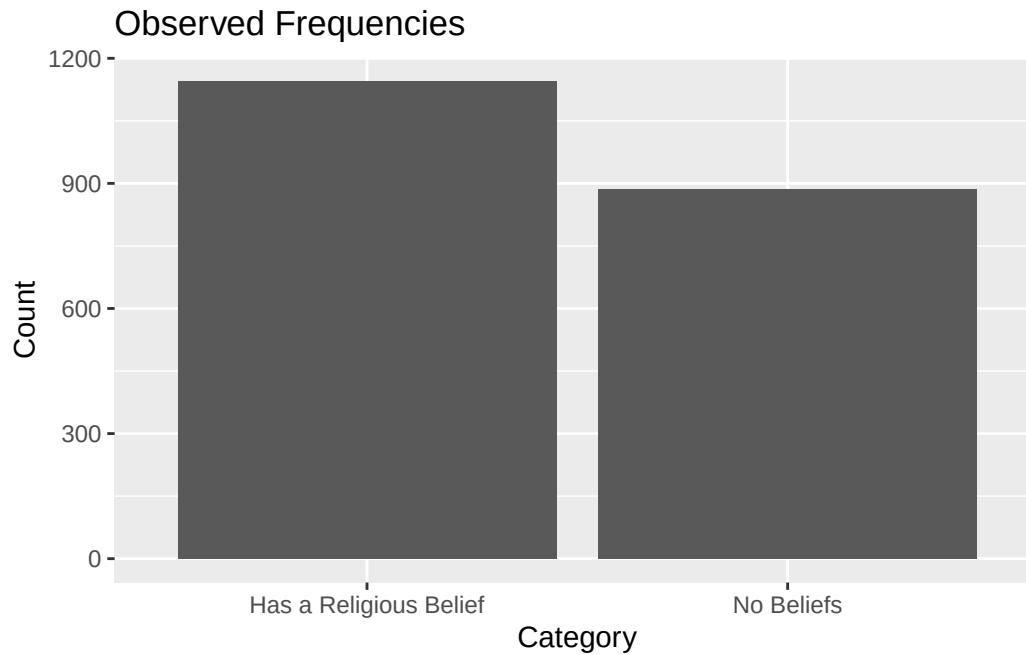
Has a Religious Belief	No Beliefs
1015.5	1015.5

```
chisq.test(table(data.xls$ONLY_R))
```

Chi-squared test for given probabilities

```
data: table(data.xls$ONLY_R)
X-squared = 33.029, df = 1, p-value = 9.082e-09
```

```
ggplot(data.xls, aes(x = ONLY_R)) +
  geom_bar() +
  labs(title = "Observed Frequencies", x = "Category", y = "Count")
```



```
table(data.xls$ONLY_R)
```

Has a Religious Belief	No Beliefs
1145	886

```
chisq.test(table(data.xls$CVD_EXAG))$expected
```

Out-of-Proportion Reasonable Concern	
1015.5	1015.5

```
chisq.test(table(data.xls$CVD_EXAG))
```

Chi-squared test for given probabilities

```
data: table(data.xls$CVD_EXAG)
X-squared = 601.19, df = 1, p-value < 2.2e-16
```

```
ggplot(data.xls, aes(x = CVD_EXAG)) +  
  geom_bar() +  
  labs(title = "Observed Frequencies", x = "Category", y = "Count")
```

